AI Agent Discovery

Presenters: Roland Schott (roland.schott@telekom.de), Nic Williams (nic@infoblox.com)

Co-Proponents: Behcet Sarikaya (sarikaya@ieee.org), Jim Mozley (jmozley@infoblox.com)

Internal

AI Discovery Current State

- "Do something new"
  - Add auth, security, etc.
- "Handle it via the network"
  - DNS, SIP-like registration with carrier networks, etc
- "Handle it via applications"
  - Model cards & well-known URL

Internal

AI Discovery Current State

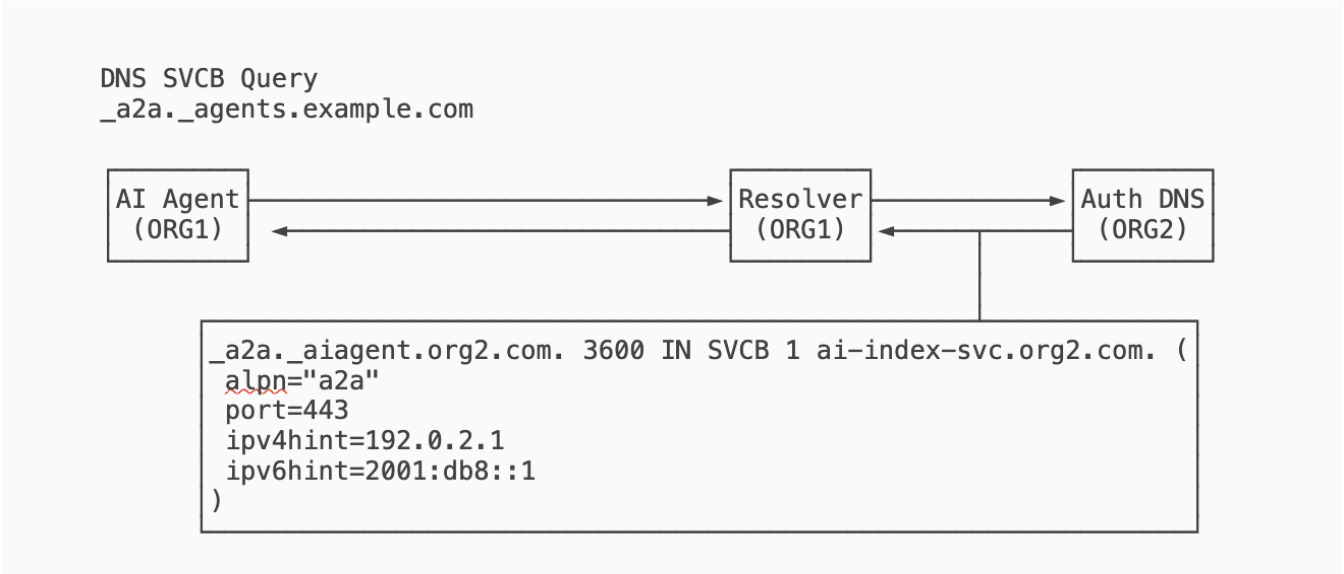
- "Do something new" -> What happens if the IETF takes too long?
  - Add auth, security, etc.
- "Handle it via the network" -> What happens if the metadata is too rich?
  - DNS, SIP-like registration with carrier networks, etc
- "Handle it via applications" -> What if application owners create dysfunction?
  - Model cards & well-known URL

AI Discovery Problem Space

- Ephemeral (dynamic de/registration), workload-specific agents could scale to billions. Discovery requests orders of magnitude more.
- Context-aware trusted discovery of offered services and cost with integration to public/private networks with operators controlling advertising their services.
- Legal dispute for agent removal or unauthorized access, legislation enforcing data sovereignty, and enforcing uniqueness / anti-collision mechanisms globally.

Internal

DNS-AID Example



SVCB params to be modified to include application-specific context (input token amounts, cost per input bundle, etc.)

Internal

DNS-AID Unanswered Questions

- Leaf attribute zones for specific capabilities (i.e. _chat._agents.example.com)?
  - Or, better for a protocol to have agents within and records hosting SVCB with capabilities in _agents.example.com where booking._agents.example.com includes ALPN or proto = A2A in the record? (i.e. chat._a2a._agents.example.com)
    - These need to be registered with IANA. What if you do something that isn't yet defined like chat/booking? Delays in processing...
    - What if you don't know what you are looking for? If you want to see all agents, does that lend credibility to an index-like service where there are well-known agent capabilities registered with IANA and innovative ones self hosted?

- AI agents typically use an API call to Google Search, or screen capture a browser UI and analyze the picture. They favor HTTPS transactions.
- How do we make the connection between 'brand' to 'domain' in a predictable manner?
  - Reputation weighting of domains or nameservers?

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